Exact Moment Coordinates, Block Mollifiers, and Uniform Rank-Tail Boundaries

A Ram Calculus Analysis of Mean Analytic Rank

Dr. Muralidhara Subbarao
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Stony Brook University
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Abstract

We study the finite-model and rank-tail interfaces in the September 23, 2026 manuscript *The mean analytic rank of quadratic twists of elliptic curves*. The source's arithmetic logarithmic moment estimates, uniform comparison theorem, and companion density theorem remain separate inputs, not conclusions of a covariance calculation. We give exact tensor moment coordinates for its integer-character model, an exact independent-block error identity retaining coherent bias, and a fixed-order Fourier energy certificate for its logarithmic weights. From explicitly uniform finite-window rank counts we prove a quantitative dyadic transfer, including the small blocks, and conditional convergence of analytic-rank moments of every fixed order $0 < q < 6/5$. A manufactured logarithmically capped family shows why the same premises do not settle $q = 6/5$. A general summable-envelope criterion and a finite-height conditional modulus specify the remaining uniform-integrability requirements. These are local deductions and conditional extensions, not an independent proof of Goldfeld's conjecture or integral BSD. Contribution assessment: C3/5, the proved source-relative fractional-moment extension and exact block-error refinement are notable, while their arithmetic and density inputs remain imported.

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1 Scope, Status, and Principal Conclusions

This report concerns catalogue entry 12, applicability **L3**, in family 6. The assigned source is OpenAI's September 23, 2026 preprint [1], pinned at commit `adc7f1241b42e322a645 1854ab7e4b4c146bf78a`. Parent-managed October 8 source-identity evidence reports unchanged bytes across all ten source texts. Byte identity is provenance, not theorem verification. The present contribution score is independently **C3**; applicability and contribution measure different things.

The source asserts mean analytic rank 1/2 for every fixed elliptic curve over $\mathbb{Q}$, ordered by signed squarefree twist parameters. Its essential extra input beyond rank-zero/one density is a rank-weighted tail estimate. We neither replace that input by a finite positive matrix nor certify the entire source proof. The major local conclusions are as follows.

1. The actual prime law has an exact three-dimensional moment space. Tensor coordinates distinguish the constant component from variance, and formal mollifier degree from orthogonal degree (Proposition 4.1).
2. A product truncation has an exact coherent-bias plus orthogonal- error formula. Under an additional zero-bias condition, squared errors accumulate through a product of $1 + v_i$, rather than a squared sum of norm errors (Theorem 5.1). The source does not automatically satisfy that extra condition.
3. The source's finite-window count and pointwise cap imply uniform integrability of a^q for every fixed $0 < q < 6/5$, and, with its companion density input, mean $a^q \to 1/2$ (Corollary 6.3). The endpoint obstruction concerns what these premises alone imply, not the actual distribution of elliptic curves.

Evidence depth. The eight assigned TeX files contain 3,138 lines. Literal bounded reads cover 2,408 distinct lines: seven files in full and 237 of 967 lines in `analytic.tex`. The bibliography was separately read in full (257 lines). Three private originals were read only in selected intervals, 324 lines in total. The source's full section and theorem structure was mapped, but the 730 unread analytic lines, companion manuscripts, and unread private passages are not claimed audited. Appendix A and the machine-readable ledger give exact scopes and hashes.

Logical vocabulary. "Source" denotes a claim or deduction in the inspected manuscript. "Local" denotes a result proved here on its stated mathematical space. "Conditional" denotes a consequence

using identified source arithmetic or external hypotheses. Finite tests check formulas and normalizations on bounded examples; they do not verify a global arithmetic theorem, a growing family uniformly in height, or historical priority.

2 Source Architecture and Normalizations

2.1 Ranks, counting, and the finite twist reduction

For a fixed elliptic curve $E/\mathbb{Q}$ let

$$\mathcal{D}(Y) = \{d \in \mathbb{Z} : 0 < |d| \le Y,\ d \text{ squarefree}\}, \qquad N(Y) = \#\mathcal{D}(Y) = \frac{2Y}{\zeta(2)} + O(\sqrt{Y}).$$

Write $a_d = \operatorname{ord}_{s=1} L(E^{(d)}, s)$ for analytic rank and $r_d = \operatorname{rank}_{\mathbb{Z}} E^{(d)}(\mathbb{Q})$ for algebraic rank. These are not silently identified. All limits fix E first; constants may depend on E and on fixed moment or derivative orders.

The source passes to a finite set $\mathcal{F}$ of twists supported on the primes S dividing $2q_E$. For $F \in \mathcal{F}$, admissible d are squarefree, $d \equiv 1 \pmod 4$, and coprime to the product Q_S of the primes in S. Here $q_{F^{(d)}} = q_F |d|^2$. Extracting the S-part of an arbitrary squarefree parameter and adjusting its sign gives an admissible parameter for some $F \in \mathcal{F}$, with height rescaled by a divisor of Q_S and bounded multiplicity. Thus nonnegative upper estimates transfer with fixed constants. This is *not* an equality of normalized probability distributions.

In the unitary normalization,

$$L_F(s) = L(F, s + \tfrac{1}{2}) = \sum_{n \ge 1} \lambda_F(n) n^{-s}, \quad C_F = \frac{\sqrt{q_F}}{2\pi}, \quad \Lambda_F(s) = C_F^s \Gamma(s + \tfrac{1}{2}) L_F(s).$$

The center is $1/2$, $|\lambda_F(n)| \le \tau(n)$, and the functional equation is $\Lambda_F(s) = \varepsilon_F \Lambda_F(1 - s)$. In local Euler factors put $\xi_p = 1$ at good primes and 0 at bad primes.

2.2 The six mathematical stages

Source stage	Mathematical role	Status in this report
Setup	Finite twists, conductor scaling, integer prime model	Full literal read; conventions retained.
Analytic	Gauss sums, Euler factors, weighted Poisson summation, logarithmic second-moment induction, polynomial comparison	Statements and final comparison read; most implementing arithmetic is imported.
Weights	Completed-derivative identity, short/terminal partition, fixed-order derivative bounds, terminal moment	Full literal read; Mellin normalizations and fixed-order boundaries checked.
Model	Finite Euler products, block truncation, integer good event, full-model norms, Fourier representation	Full literal read; local exact-coordinate reformulations proved below.
Tail	Uniform k^{-2} count, logarithmic rank cap, dyadic tail, mean	Full literal read; conditional fractional extension below.
Algebraic	Exponential algebraic moments plus density-one rank equality	Full literal read; external moment theorem remains imported.

2.3 What the logarithmic moment input supplies

The source's analytic propositions assert fixed exponents A, B for dyadic family means of $|L_{F^{(d)}}(\sigma + it)|^2$, bounded by a power of $\log M$ times $(1+|t|)^A$, uniformly for $1/2 \le \sigma \le 2$. The short-polynomial version is

$$\frac{1}{X} \sum_{d \in \mathcal{A}(X)} \left| \sum_n \frac{\lambda_F(n)\chi_d(n)}{\sqrt{n}} n^{-it} G(n/N) \right|^2 \ll_G \ell^B (1+|t|)^A (1+N/X)^A, \qquad \ell = \log X, \tag{1}$$

for fixed smooth compactly supported G and admissible $\mathcal{A}(X) = \{d : X \le |d| < 2X\}$. This family norm has mass $\#\mathcal{A}(X)/X$, not necessarily one. The word "logarithmic" here includes a logarithmic *loss*; it does not turn an arithmetic second moment into a continuous Mellin Parseval identity.

For a short weight of length $X^{1-\Delta}$ and a mollifier $H_V = \sum_{r \le R_0} h(r)V(r)$, the comparison in `analytic.tex:807-967` requires

$$c_1 \ell^{-1/10} \le \Delta \le \tfrac{1}{2}, \quad R_0 \le X^{\eta\Delta}, \quad \sum |h(r)| \le C_H R_0^2,$$

and fixed-order scaled derivative seminorms bounded by powers of ℓ. With appropriate fixed smoothing and deterministic A_x, its form is

$$\frac{1}{X} \sum_m \Psi(m/X) |H_{z_m} S_{z_m}(m/X) - A_{m/X}|^2 = \int \Psi(x) \mathbb{E} |H_Z S_Z(x) - A_x|^2 \, dx + O(e^{-c\Delta\ell}). \tag{2}$$

The sum is the source's smoothed integer family in its fixed local class, with its corresponding zero-frequency normalization. The model fixes the value at 2 and uses independent odd primes as below. One restricts a nonnegative integer sum to admissible squarefree parameters for an upper bound; one does not replace the zero-frequency law by the squarefree law. The inspected proof absorbs a factor of $R_0^{4+2\cdot 257}$ into $X^{-\Delta}$ by choosing $\eta = 1/[4(4+2\cdot 257)]$. Its earlier nonzero-frequency estimates, not finite Gram positivity, carry the arithmetic content.

3 Completed Derivatives and Logarithmic Energy

3.1 The source's exact vanishing detector

Fix derivative orders first, then a sufficiently large fixed k_0, then one height threshold valid for

$$k_0 \le k \le \ell^{3/5}, \quad J = \left\lfloor \frac{\log \ell}{10 \log 2} \right\rfloor, \quad \delta_j = 2^{-j}, \quad s = k/\ell.$$

In particular $\delta_J \asymp \ell^{-1/10}$ and $k/(\ell\delta_j) \ll \ell^{-3/10}$. No derivative order grows with k. For x in a fixed compact set away from zero the exact weight is

$$W_{k,x}(u) = \int_0^\infty e^{-v} \left(1 - u + \frac{\log(C_F|x|v)}{\ell}\right)_+^k dv.$$

With $g_d(z) = L_{F^{(d)}}(1/2+z)(C_F|d|)^z \Gamma(1+z)$ and $S_d = \sum_n \lambda_F(n)\chi_d(n) n^{-1/2} W_{k,d/X}(\log n/\ell)$, the source proves

$$(1 + \varepsilon_d(-1)^k) S_d = \ell^{-k} g_d^{(k)}(0). \tag{3}$$

Thus $a_d > k$ forces $S_d = 0$ only for the matching parity $\varepsilon_d = (-1)^k$. The opposite sign requires $k-1$; dividing by the prefactor when it is zero would invalidate the argument.

The smooth partition gives $W = w_0 + \sum_{j=1}^{J} w_j + w_*$, with short length $X^{1-\delta_j/4}$ and terminal exponential decay in n/X. The main deterministic term is $A_x = (C_F|x|)^s\Gamma(1+s)$, bounded above and below. The source estimates in the full model are

$$\|H_0 S_0 - A_x\|_2 \ll k^{-1}, \qquad \|H_j S_j\|_2 \ll (k2^j)^{C_5}\delta_j^k. \tag{4}$$

The terminal family mean square is $\ll \ell^{C_T}\delta_J^{2k}$, with C_T fixed *before* k_0. Taking $k_0 > 20C_T$ absorbs this into $2^{-3Jk/2}$ uniformly in the entire k-window. Similarly $e^{-c\Delta_j\ell} \le k^{-L}2^{-Ljk}$ for any fixed L, because $\Delta_j\ell \gg \ell^{9/10}$ whereas $jk \ll \ell^{3/5}\log\ell$.

3.2 Exact Mellin normalization from the private proof route

The inspected Mellin proof in the private volume-2 chapter establishes the following classical identity. With $\widehat{g}(t) = \int_{\mathbb{R}} g(y)e^{-ity}\,dy$ and $\mathcal{M}f(z) = \int_0^\infty f(x)x^{z-1}\,dx$, let $g_c(y) = e^{cy}f(e^y) \in L^1(\mathbb{R}) \cap L^2(\mathbb{R})$. Then

$$\mathcal{M}f(c+it) = \widehat{g}_c(-t), \qquad \int_0^\infty |f(x)|^2 x^{2c-1}\,dx = \frac{1}{2\pi}\int_{\mathbb{R}} |\mathcal{M}f(c+it)|^2\,dt. \tag{5}$$

The L^2 extension follows by density. Inversion is in L^2, or pointwise under the usual stronger inversion hypotheses, not at every point of an arbitrary L^2 representative. The source terminal transform uses $c = 0$ and measure dy/y. In contrast $c = 1/2$ gives dx. If $f(x) = h(\log x/\ell)$, then

$$\int |f(x)|^2\frac{dx}{x} = \ell\int |h(u)|^2 du, \qquad \mathcal{M}f(it) = \ell\widehat{h}(-\ell t).$$

The ℓ and 2π factors are compulsory. A discrete $n^{-1/2}$ coefficient is not a continuous change-of-variable Jacobian.

Proposition 3.1 (Fixed-order Fourier energy certificate)**.** *Let $\mathcal{H}$ be a complex Hilbert space, $V(t)$ strongly measurable, and $\|V(t)\|_{\mathcal{H}} \le C(1+t^2)^{b/2}$ for $b \ge 0$. For an integer $m > b + 1/2$ and $g \in H^m(\mathbb{R})$ put*

$$\mathcal{E}_m(g) = \sum_{r=0}^{m}\binom{m}{r}\left\|g^{(r)}\right\|_{L^2}^2, \qquad I_{m-b} = \int_{\mathbb{R}}(1+t^2)^{-(m-b)}dt < \infty.$$

Then the Bochner integral below exists and

$$\left\|\frac{1}{2\pi}\int_{\mathbb{R}}\widehat{g}(t)V(t)dt\right\|_{\mathcal{H}} \le \frac{C}{\sqrt{2\pi}}\sqrt{I_{m-b}\mathcal{E}_m(g)}. \tag{6}$$

Proof. Minkowski bounds the norm by $(C/2\pi)\int|\widehat{g}(t)|(1+t^2)^{b/2}dt$. Cauchy–Schwarz with factors $(1+t^2)^{m/2}$ and $(1+t^2)^{-(m-b)/2}$ gives the stated integral constant. Plancherel and the binomial theorem give $\int(1+t^2)^m|\widehat{g}(t)|^2dt = 2\pi\mathcal{E}_m(g)$. The same bound proves absolute Bochner integrability, justifying the argument by truncated integrals if necessary. □

For the source's first-weight Fourier error, $g_x(y) = e^{y/2}(w_{0,x}(y/k) - A_x e^{-y})$ after its smooth extension on $[-2, 0]$. The inspected global truncated-power argument and its exponential majorants give fixed-order H^m energy $O_m(k^{-2})$, uniformly in x in the fixed compact set. The source polynomial shift bounds supply $V(t)$ in the full model L^2 space, uniformly for finite Euler cutoff Q. Proposition 3.1 therefore reproduces the $O(k^{-1})$ error in (4) with an explicit energy budget. This is an alternative certificate, not an improved arithmetic exponent. A numerical approximation to g still needs a proved H^m residual bound; sampled values alone supply none. For the terminal family estimate the kernel depends on d/X. One must retain its uniform-in-x bound before the family norm, as the source does. A scalar Parseval identity cannot remove that supremum.

4 Exact Finite Ram Moment Coordinates

The private finite-energy and joint-moment proofs motivate exact coordinates with their actual metric. We use them on the source's specific finite probability space, not on an invented Gaussian model. For a finite set P of odd primes, the variables Z_p are independent, with

$$\mathbb{P}(Z_p = 0) = 1/p, \qquad \mathbb{P}(Z_p = 1) = \mathbb{P}(Z_p = -1) = \tfrac{1}{2}(1 - 1/p). \tag{7}$$

The value $Z_2 = c \in \{-1, 1\}$ is deterministic and absorbed into coefficients. This is the integer-character model coming from complete residue classes. It is not the squarefree model, whose local zero probability would instead be $1/(p+1)$.

Proposition 4.1 (Exact ternary tensor metric)**.** *Put $q_p = 1 - 1/p$. On $L^2(\{-1, 0, 1\}, \mathbb{P}_p)$, the raw basis $(1, z, z^2)$ has Gram matrix*

$$G_p = \begin{pmatrix} 1 & 0 & q_p \\ 0 & q_p & 0 \\ q_p & 0 & q_p \end{pmatrix}, \qquad \det G_p = q_p^2(1 - q_p) > 0.$$

The basis $\psi_{p,0} = 1$, $\psi_{p,1} = z$, $\psi_{p,2} = z^2 - q_p$ is orthogonal, with squared norms $d_{p,0} = 1$, $d_{p,1} = q_p$, $d_{p,2} = q_p(1 - q_p)$. Every complex function F on the finite product support has a unique expansion

$$F(Z) = \sum_{\alpha \in \{0,1,2\}^P} c_\alpha \prod_{p \in P} \psi_{p,\alpha_p}(Z_p), \qquad \mathbb{E}|F|^2 = \sum_\alpha |c_\alpha|^2 \prod_p d_{p,\alpha_p}.$$

Its mean is c_0 and its variance is the same sum with $\alpha = 0$ omitted. Equivalently, raw coefficients have tensor Gram metric $\bigotimes_p G_p$.

Proof. Odd positive moments vanish; every even positive moment is q_p. Direct calculation gives the displayed matrices and orthogonality. The three basis functions are linearly independent on three points of positive mass. Tensoring their bases and inner products proves completeness and the norm formula by independence. Complex coefficients require conjugation in the inner product, hence squared moduli. □

Application to the actual mollifier. For $s = k/\ell$, $Y_j = X^{\delta_j^2/\sqrt{k}}$, the source splits primes into $I_{j,i} = (Y_j^{2^{-i-1}}, Y_j^{2^{-i}}]$ and forms

$$K_{j,i}(Z, z) = \prod_{p \in I_{j,i}} (1 - \lambda_F(p) Z_p p^{-1/2-s} z + \xi_p Z_p^2 p^{-1-2s} z^2).$$

It retains formal z-degree at most $m_{j,i} = \lceil k^{1/4} + j + i \rceil$, evaluates at $z = 1$, and multiplies these truncated blocks to obtain H_j. First perform this formal truncation; only afterwards reduce functions using $Z_p^3 = Z_p$ and transform into Proposition 4.1's coordinates. Formal degree is not orthogonal degree. In particular the discarded term uZ_p^2 has mean uq_p and energy $|u|^2 q_p$, not zero mean or energy $|u|^2 q_p(1 - q_p)$. This elementary mismatch can otherwise erase precisely the coherent error a mollifier must control.

The source's length bound is

$$\log R_j \le C_{\text{len}}(k^{1/4} + j + 1)\frac{\ell \delta_j^2}{\sqrt{k}},$$

which meets $R_j \le X^{\eta \delta_j/4}$ after choosing fixed k_0 large enough. Exact tensor coordinates do not shorten this support. Direct enumeration costs $3^{|P|}$; sparsity or tensor contraction can reduce particular computations, but no general complexity or speedup claim is made.

Probability and arithmetic are distinct. Gram positivity proves a norm identity for the common joint law. The private covariance remainder result requires an actual L^2 error. Neither a positive matrix nor a small probability of a bad event implies a small rank-weighted or energy-weighted tail. The source's full-model L^2 estimates precede restriction to its integer good event. Conditioning on that event can destroy independence and cannot be inserted into the tensor proof.

5 A Sharper Independent-Block Error Identity

The source's relative truncation lemma uses a telescoping norm bound. The following local identity retains more information at the same finite-product level.

Theorem 5.1 (Bias and orthogonal block errors)**.** *Let (A_i, B_i), $1 \le i \le n$, be independent pairs of complex L^2 random variables, each on its own block probability space. Suppose $a_i = \mathbb{E}|A_i|^2 > 0$ and define*

$$b_i = \frac{\mathbb{E}(B_i\overline{A_i})}{a_i}, \qquad v_i = \frac{\mathbb{E}|B_i|^2}{a_i} - |b_i|^2 \ge 0.$$

Then

$$\frac{\mathbb{E}|\prod_i B_i - \prod_i A_i|^2}{\prod_i a_i} = \left|\prod_i b_i - 1\right|^2 + \prod_i (|b_i|^2 + v_i) - \prod_i |b_i|^2. \tag{8}$$

If $b_i = 1$ for every i, the right side is $\prod_i(1+v_i) - 1 \le \exp(\sum_i v_i) - 1$. More generally, if $|b_i - 1| \le \beta_i$ and $v_i \le \nu_i$, it is at most

$$\left(\prod_i (1+\beta_i) - 1\right)^2 + \prod_i ((1+\beta_i)^2 + \nu_i) - \prod_i (1+\beta_i)^2. \tag{9}$$

Proof. Writing $B_i = b_i A_i + D_i$ gives $\mathbb{E}(D_i\overline{A_i}) = 0$ and $\mathbb{E}|D_i|^2 = a_i v_i$. Alternatively expand the left side of (8); independence gives $\prod_i(|b_i|^2 + v_i) + 1 - 2\Re\prod_i b_i$. Adding and subtracting $\prod_i |b_i|^2$ proves the identity. For $b_i = 1$, use $1 + v_i \le e^{v_i}$. In general a telescoping expansion bounds $|\prod b_i - 1|$ by $\prod(1+\beta_i) - 1$. The remaining difference is a polynomial with nonnegative coefficients in $|b_i|^2$ and v_i, since its empty-subset term cancels. Replacing these variables by their upper bounds proves (9). □

For the source take $A_i = K_{j,i}(Z, 1)L_{I_{j,i}}(b)$ and $B_i = T_{j,i}(Z)L_{I_{j,i}}(b)$, where b is the complex shift and $L_I(b)$ is the finite Euler product on that block. The source verifies nonvanishing denominators for $\Re b \ge 0$. On a finite ternary support these are finite functions and hence belong to the exact space of Proposition 4.1, even when represented as rational functions. An independent residual product outside Y_j multiplies both squared norms by the same positive factor and cancels. No infinite-prime limit is needed for (8); any use as $Q \to \infty$ would require separate convergence.

The source bounds $\|B_i - A_i\|_2 \le \epsilon_i \|A_i\|_2$ with $\epsilon_i \ll 2^{-m_{j,i}}$, leading to a relative product norm at most $\exp(\sum_i \epsilon_i) - 1$. The exact identity is always available, but improvement depends on the actual b_i, v_i. For n blocks with $b_i = 1$ and $v_i = \epsilon^2$, its squared bound is $(1+\epsilon^2)^n - 1 \sim n\epsilon^2$, whereas the squared telescoping bound is $((1+\epsilon)^n - 1)^2 \sim n^2\epsilon^2$ for fixed n as $\epsilon \to 0$. This is a genuine conditional local improvement. With coherent bias $B_i = (1+\epsilon)A_i$, $v_i = 0$ and $b_i = 1 + \epsilon$, the exact formula instead retains the full squared telescoping expression. Formal truncation does not prove $b_i = 1$; we claim no improved source rank exponent from it.

5.1 Residuals and comparison errors have different units

Proposition 5.2 (A second-moment error budget)**.** *On any finite measure space, if $F = \widetilde{F} + R$, $\|R\|_2 \le \epsilon$, and $\left\|\widetilde{F}\right\|_2^2 = \mu$, then*

$$(\max\{0, \sqrt{\mu} - \epsilon\})^2 \le \|F\|_2^2 \le (\sqrt{\mu} + \epsilon)^2.$$

If an arithmetic nonnegative second moment differs from $\|F\|_2^2$ by at most ρ, its upper bound is $(\sqrt{\mu} + \epsilon)^2 + \rho$. Its measure of the event $|F_{\mathrm{arith}}| > t$ is at most that quantity divided by t^2.

Proof. The two norm bounds are the triangle and reverse triangle inequalities. Add the comparison error and apply Chebyshev's inequality on the arithmetic measure space. □

For the main detector the threshold is bounded away from zero on the source good event. Preserving its k^{-2} count therefore requires a main model norm $O(k^{-1})$, a norm residual $O(k^{-1})$, and an additive second-moment error $O(k^{-2})$. Later-block thresholds depend on j, k and must be divided out *before* summing errors. A height-dependent approximation with a fixed nonzero error floor cannot yield the required uniform tail. Discarding an event of probability ϵ costs its *energy* on that event, not merely ϵ.

6 Finite-Window Tails and Fractional Rank Moments

6.1 An exact finite layer-cake interface

Theorem 6.1 (Finite-window moment tails)**.** *Let a be a nonnegative integer-valued variable on a probability space, $a \le U$, where $U \ge K \ge R \ge k_* \ge 1$ are integers. Set $T(k) = \mathbb{P}(a > k)$ and $\Delta_q(k) = (k+1)^q - k^q$ for $q > 0$. Then*

$$\mathbb{E}[a^q \mathbf{1}_{a>R}] \le R^q T(R) + \sum_{k=R}^{K-1} \Delta_q(k) T(k) + (U^q - K^q) T(K). \tag{10}$$

If $T(k) \le Ck^{-p}$ for every integer $k_* \le k \le K$, where $p > q > 0$, then

$$\mathbb{E}[a^q \mathbf{1}_{a>R}] \le CC_{p,q} R^{q-p} + CU^q K^{-p}, \tag{11}$$

with a constant depending only on p, q. If $T(k) \le Ck^{-p} + e(k)$ instead, add the nonnegative budget

$$R^q e(R) + \sum_{k=R}^{K-1} \Delta_q(k) e(k) + U^q e(K).$$

Proof. For each integer $a > R$, $a^q = R^q + \sum_{k=R}^{a-1} \Delta_q(k)$. Averaging gives equality with upper summation limit $U - 1$. For $k \ge K$, use $T(k) \le T(K)$ and telescope to obtain (10). For $k \ge 1$, the mean value theorem gives $\Delta_q(k) \le q \max(1, 2^{q-1}) k^{q-1}$. The resulting convergent series $\sum_{k \ge R} k^{q-p-1}$ is $O_{p,q}(R^{q-p})$. The error assertion follows from the same nonnegative sums. □

In a varying family the sufficient upper-window condition is $U_X^q / K_X^p \to 0$, not merely $K_X \to \infty$. If $U_X \ll (\log X)^\beta$ and $K_X \asymp (\log X)^\theta$, the guaranteed range is

$$0 < q < \min\{p, \theta p/\beta\} \qquad (\beta > 0). \tag{12}$$

For the source, $p = 2$, $\theta = 3/5$, $\beta = 1$. It is the uncounted window from K_X to U_X, rather than the formal summability threshold $q < 2$, that limits this argument.

6.2 A quantitative dyadic transfer with explicit small blocks

The source count proposition and Jensen cap, together with its finite twist reduction, supply the following inputs *if accepted*. There are fixed $C, c, X_0 > 0$ and an integer k_* such that, for every real $X \geq X_0$ and every integer $k_* \leq k \leq \lfloor c(\log X)^{3/5} \rfloor$,

$$\#\{d \in \mathcal{D}(2X) \setminus \mathcal{D}(X) : a_d > k\} \leq CXk^{-2}, \qquad a_d \leq C \log(3|d|). \tag{13}$$

Endpoint conventions in the dyadic intervals change at most a bounded number of parameters and can be absorbed in constants. Fixed rescalings by divisors of Q_S shrink c but do not change the exponent; a uniform count for the finite set $\mathcal{F}$ is required. These are explicit uniform hypotheses, stronger information than a bare fixed-R limsup conclusion.

Theorem 6.2 (Uniform cumulative bound)**.** *Assume* (13)*. Fix* $0 < q < 2$*. For all sufficiently large* Y *and integers*

$$k_* \leq R \leq \left\lfloor c(\tfrac{1}{2}\log Y)^{3/5} \right\rfloor,$$

one has, uniformly in this R*-window,*

$$\frac{1}{N(Y)} \sum_{d \in \mathcal{D}(Y)} a_d^q \mathbf{1}_{a_d > R} \ll_{E,q} R^{q-2} + (\log Y)^{q-6/5} + Y^{-1/2}(\log Y)^q. \tag{14}$$

Proof. Partition the upper part of $\mathcal{D}(Y)$ into dyadic blocks of height $X = Y/2^{j+1} \geq \sqrt{Y}$. Their total heights are at most Y; the unpartitioned part has $|d| \leq 2\sqrt{Y}$. In each large block $K_X = \lfloor c(\log X)^{3/5} \rfloor \geq R$ and $U_X = \lceil C \log(6X) \rceil$. The counting-measure version of Theorem 6.1 (multiply its proof by the block population) gives

$$\sum_{X < |d| \leq 2X} a_d^q \mathbf{1}_{a_d > R} \ll X\{R^{q-2} + (\log X)^{q-6/5}\}.$$

Since $\frac{1}{2}\log Y \leq \log X \leq \log Y$, this is uniformly $\ll_q X\{R^{q-2} + (\log Y)^{q-6/5}\}$, for either sign of $q - 6/5$. In the remaining small part there are $O(\sqrt{Y})$ parameters, each of rank $O(\log Y)$, so its entire q-moment is $O(\sqrt{Y}(\log Y)^q)$. Divide by $N(Y) \asymp Y$. This proof has no unspecified R-dependent small-block constant. □

Corollary 6.3 (Conditional fractional analytic-rank extension)**.** *Assume* (13)*. For every fixed* $0 < q < 6/5$*,*

$$\lim_{R \to \infty} \limsup_{Y \to \infty} \frac{1}{N(Y)} \sum_{d \in \mathcal{D}(Y)} a_d^q \mathbf{1}_{a_d > R} = 0.$$

If, in addition, the companion density assertion $\#\{d \in \mathcal{D}(Y) : a_d = j\}/N(Y) \to 1/2$ *holds for* $j = 0, 1$*, then*

$$\frac{1}{N(Y)} \sum_{d \in \mathcal{D}(Y)} a_d^q \longrightarrow \frac{1}{2} \qquad \textit{for every fixed } 0 < q < 6/5. \tag{15}$$

Proof. For fixed R, the last two terms in (14) vanish when $q < 6/5$; then $R^{q-2} \to 0$. On $2 \leq a_d \leq R$ the normalized contribution is at most $R^q \mathbb{P}_Y(a_d \geq 2)$, which vanishes by the density premise. Rank zero contributes zero, rank one contributes its density, and the tail is uniformly integrable. Take $Y \to \infty$ at fixed R before $R \to \infty$. □

The interval $1 < q < 6/5$ extends the source's stated analytic mean conclusion using its stronger finite-height estimates. It does not independently establish those estimates or the companion density theorem, and it asserts neither a second analytic-rank moment nor analytic exponential moments.

6.3 The endpoint is not a consequence of these bounds

Example 6.4 (A logarithmically capped endpoint obstruction)**.** For integers $L \geq 2$, take a finite population of size $N_L = 2L^6 2^{L^5}$. Assign ranks zero and one to $(L^6 - 1)2^{L^5}$ members each, and rank L^5 to the remaining 2^{L^5+1} members. These are artificial integer-valued data, not elliptic curves. Put $K_L = L^3$ and $U_L = L^5$. For $1 \leq k \leq K_L$,

$$T_L(k) = L^{-6} \leq k^{-2}, \quad K_L \asymp (\log N_L)^{3/5}, \quad U_L \asymp \log N_L.$$

Both rank-zero and rank-one densities tend to $1/2$, and

$$\mathbb{E}_L a^q = \tfrac{1}{2}(1 - L^{-6}) + L^{5q-6}.$$

Thus the mean tends to $1/2$, but the $6/5$ moment tends to $3/2$, with a nonvanishing tail of mass-weighted size one. For $q > 6/5$ these moments diverge. All computations are exact.

This triangular-family construction proves only insufficiency of the finite-window count and cap for the endpoint. It is not a counterexample to any elliptic-curve conjecture. In particular, even a logarithmic pointwise rank cap does not automatically upgrade convergence in distribution to convergence of moments. The parent has separately recorded a different manufactured example where the mean itself fails to follow from density; it is not counted as an author test here.

7 Summable Envelopes and Conditional Finite-Height Moduli

Proposition 7.1 (A summable-envelope alternative)**.** *Let $a_X \leq U_X$ be nonnegative integer-valued on probability spaces. Suppose $K_X \to \infty$ and $T_X(k) \leq H(k)$ for $k_* \leq k \leq K_X$, where H is nonnegative, nonincreasing and $\sum_{k \geq k_*} H(k) < \infty$. If $U_X H(K_X) \to 0$, then $\{a_X\}$ is asymptotically uniformly integrable:*

$$\lim_{R \to \infty} \limsup_{X \to \infty} \mathbb{E}[a_X \mathbf{1}_{a_X > R}] = 0.$$

Proof. The $q = 1$ identity gives the bound $RH(R) + \sum_{k=R}^{K_X - 1} H(k) + U_X H(K_X)$. The series tail vanishes. Monotonicity also gives $(R/2)H(R) \leq \sum_{\lfloor R/2 \rfloor \leq k \leq R} H(k)$ up to harmless integer endpoints, so $RH(R) \to 0$. □

For instance $H(k) = C/[k(\log(e + k))^{1+\eta}]$, $\eta > 0$, is summable, so a power exponent strictly larger than one is not logically necessary. But with the source's $U_X \asymp \log X$ and $K_X \asymp (\log X)^{3/5}$, this particular weaker envelope would not make $U_X H(K_X)$ vanish. One must improve the window or cap, or control the remaining tail separately. Summability alone does not settle the unobserved upper window.

Corollary 7.2 (A finite-height modulus under uniform premises)**.** *Assume* (13) *and define actual finite-height errors*

$$\alpha_Y = \left| \mathbb{P}_Y(a = 1) - \tfrac{1}{2} \right|, \qquad \delta_Y = \mathbb{P}_Y(a \geq 2), \qquad B_q(Y) = (\log Y)^{q-6/5} + Y^{-1/2}(\log Y)^q.$$

For $0 < q < 6/5$ and every integer R in Theorem 6.2's window,

$$\left| \mathbb{E}_Y a^q - \tfrac{1}{2} \right| \leq \alpha_Y + R^q \delta_Y + C_{E,q} R^{q-2} + C_{E,q} B_q(Y). \tag{16}$$

If $\delta_Y > 0$ and $R = \lceil \delta_Y^{-1/2} \rceil$ belongs to that window, this is $\leq \alpha_Y + O_{E,q}(\delta_Y^{1-q/2} + B_q(Y))$.

Proof. Separate rank one, $2 \le a \le R$, and $a > R$. The middle part is bounded by $R^q \delta_Y$ and the last by (14). For the displayed choice, $R \asymp \delta_Y^{-1/2}$ since $0 < \delta_Y \le 1$, and both powers have the stated size. If that choice is outside the window, use (16) at a permissible integer instead. □

For $q = 1$ this gives the conditional $\sqrt{\delta_Y}$ modulus plus the source-sized $(\log Y)^{-1/5}$ remainder and the explicit small-block term. If $\delta_Y = 0$, the mean error is simply α_Y. This is not a newly proved density rate: the source's density assertion is qualitative. Optimizing $R = R(Y)$ in its fixed-R limsup theorem, which contains a small-block constant $B_{E,R}$, would not be justified. Corollary 7.2 instead uses the explicitly uniform finite-window premises and the separate proof of Theorem 6.2.

8 Current Literature and the Algebraic Boundary

The public literature refresh was performed on October 8, 2026, using public titles and public URLs only. No private title, proof text, or local path was sent to a search service. The bounded search is not an exhaustive novelty review. The following primary-source distinctions matter for this report.

Smith's 2025 preprint [3] establishes the zero/one 2^∞-Selmer-corank distribution and a BSD-conditional route to Goldfeld. Selmer corank is not, without further input, analytic rank. The source's companion analytic density preprint [2] is an imported claim here, not independently certified by that literature search. Fiorilli's work [8] gives mean-rank conclusions under a hypothesis stronger than the elliptic-curve Riemann hypothesis; that hypothesis is not generated by a finite Gram identity.

Li's second-moment work [5] supplies a primary analytic benchmark; its published second-moment theorem does not by itself verify every uniform weighted comparison in the present source. Kowalski–Michel–VanderKam [6] study nonvanishing of high completed derivatives in the prime-level modular-form family. That is important precedent for derivative mollification, but the family and uniformity problem differ. Radziwiłł–Soundararajan [7] study central-value moments and distribution, not the rank-weighted tail at issue here.

The refresh also located Nathan Creighton's article [9], published October 7, 2026. Its abstract and Theorem 1.2 concern upper tails of logarithms of central values in an even-sign twist family, with block/mollifier methods. The reported improvement concerns large central values. It is not an estimate for high orders of vanishing, and its fractional central-value moments are not the fractional *rank* moments in Corollary 6.3. Only its statement and methodological context were inspected, not its full proof.

8.1 Why algebraic exponential moments do not close this problem

Koymans–Smith [4], Theorem 1.4, was checked in the primary public HTML statement. For a fixed abelian variety over a number field and polynomially parameterized quadratic twists, it bounds fixed exponential moments of *algebraic* rank over integer boxes, with the height threshold allowed to depend on the exponential parameter. The source specializes to $F = \mathbb{Q}$, $A = E$, $P(u) = u$:

$$\frac{1}{2Y} \sum_{0<|v|\le Y} e^{\kappa r(E^{(v)})} \le \exp\exp(C_E \kappa)$$

for each fixed $\kappa > 0$ and sufficiently large Y. Restricting a nonnegative sum to signed squarefree parameters introduces the limiting normalization factor $\zeta(2)$. With density-one equality of analytic and algebraic ranks and the zero/one density input, Cauchy–Schwarz controls the exceptional algebraic contribution using a fixed 2κ moment. This gives the source's algebraic exponential and integer power moments.

That argument does not bound the analytic rank on the density-zero exceptional set. Replacing r_d by a_d inside the exponential estimate would assume precisely information not supplied. We therefore retain two separate routes: imported algebraic exponential moments for r_d, and the uniform analytic finite-window counts for a_d. Neither implies the full integral BSD formula through these local norm manipulations.

9 Benefits, Limitations, and Research Directions

Demonstrated benefits. The exact ternary metric exposes errors caused by the wrong local law, by discarding a constant component, and by confusing formal truncation with orthogonal projection. The block identity is an exact refinement of a particular source step, and can improve a bound when the computed coherent bias is small. The Fourier energy certificate keeps the logarithmic scale and normalization explicit. Most importantly, the uniform cumulative bound yields a conditional range of fractional analytic-rank moments and an explicit finite-height error interface, while accounting for small blocks.

Costs and limitations. Exact prime-state enumeration grows exponentially. Bounds uniform in prime cutoff, shift and height still need analytic estimates. Neither positive covariance nor tiny unweighted exceptional density controls the arithmetic tail. No improved k-window, improved arithmetic second-moment exponent, numerical rank statistics, or algorithmic speedup has been established here. The pointwise rank cap and the density companion remain source inputs. The inspected analytic comparison proof depends on 730 source lines not read in this report and on external arithmetic results not reproved here. Thus this report is not a complete proof audit, and makes no unconditional new Goldfeld, BSD, or RH claim.

Specific future work. First, compute and rigorously bound the actual block correlations b_i and residuals v_i across the source's allowed shifts, while preserving its length restriction; a useful target is a provable separation of coherent bias from the geometric truncation norm. Second, audit the full prime-square inflation and weighted Poisson induction, including dependence on every fixed seminorm. Third, improving the admissible derivative window from $\ell^{3/5}$ to ℓ^{θ} would, with the same count and cap, extend the guaranteed moment range to $q < \min(2, 2\theta)$; this is a research target, not a claimed extension of that window. Fourth, endpoint control requires a genuine bound for ranks beyond K_X or a better cap; Example 6.4 explains why coordinate changes alone cannot supply it. Finally, any claimed effective rate should include an actual density error and explicit arithmetic constants, not an optimized qualitative limsup.

Contribution assessment. Under the project's separate five-point scale, **C3/5** is supported by Theorems 5.1 and 6.2, Corollaries 6.3 and 7.2, and the endpoint obstruction. The contribution is notable relative to the source: an exact local error refinement and a proved conditional analytic-rank moment extension. Classical Hilbert-space, Mellin, and layer-cake principles are not claimed new. No credit is assigned to inherited source theorems, prospective improvements, or the length of this report. The remaining arithmetic boundary rules out a higher claim of a new global solution.

A Literal-Read and Proof-Evidence Map

The main file and bibliography lie in the assigned build directory; all other files in the following table lie in its `sections/` subdirectory. Intervals are inclusive and describe literal untruncated

outputs, not merely searches for theorem names.

File	Literal intervals	Lines
`main.tex`	1–62	62
`introduction.tex`	1–165, 166–275	275
`setup.tex`	1–181	181
`analytic.tex`	1–76, 807–870, 871–967	237
`weights.tex`	1–180, 181–340, 341–503	503
`model.tex`	1–219, 220–349, 350–477, 478–604, 605–734, 735–824	824
`tail.tex`	1–165, 166–247	247
`algebraic.tex`	1–79	79
Assigned TeX total	7 full files, 1 partial	2,408
`references.bib`	1–155, 156–257; ancillary bibliography	257

The uninspected analytic interval 77–806 contains normalized Gauss sums, Euler-factor and weighted-Poisson calculations, most of the logarithmic induction, and short-polynomial frequency estimates. Its headings were mapped, but its proofs remain imported.

The private routes are identified by stable local identifiers to avoid unnecessary reproduction of confidential manuscript titles. The exact absolute paths and full SHA-256 hashes are retained in `_audit/M012-read-scope.json`.

Private original	Literal intervals	Used proof content
RamCalBooks, volume 2, chapter 01	425–511 (87 lines of 2,088)	Mellin logarithmic change of variables and energy; incidental nearby text is counted but not used.
TR1674 revised v2	168–218, 266–361 (147 of 515)	Energy/Gram metric, finite formulation and logical boundaries. The quarter-scale proof is not claimed read.
Core pattern 16	216–255, 305–354 (90 of 939)	Joint-moment positivity, actual L^2 remainder and covariance boundary.

No other private corpus reading is asserted. An initially truncated locator scan received no proof-read credit; the recorded passages were subsequently displayed in bounded complete outputs. Source and private scientific hashes were recorded before the respective reads and rechecked at freeze. The bibliography's hash was computed before reading but its first display was suppressed by shell table formatting; the visible value was obtained between its two chunks, and this timing distinction is retained in the ledger.

B Verification Contract

The companion author-owned test script checks bounded exact examples of the ternary Gram metric, formal-degree bias, CRT integer law, independent block identity and its coherent-bias case, finite

layer-cake inequalities, source exponents, endpoint construction, and summable-envelope boundary. It also checks the five author lines and the abstract's final contribution sentence. These tests support the displayed proofs; they are not a test of elliptic curves, arithmetic Poisson estimates, all heights, or an external theorem.

The final evidence package records native editor/compiler attempts, the requested exported PDF build, its full-page visual inspection, all actual source/private read intervals and hashes, the independent contribution assessment, and a freeze manifest. Shared parent evidence is acknowledged but not counted as author tests or frozen as mutable author-owned data. Compilation and visual QA establish artifact integrity, not proof correctness.

References

[1] OpenAI, *The mean analytic rank of quadratic twists of elliptic curves*, September 23, 2026. Pinned public source listed in Section 1; publisher manuscript.

[2] OpenAI, *Goldfeld's analytic density conjecture and the 2-converse for elliptic curves*, September 23, 2026, Theorem 1.2 as cited by the assigned source. Publisher manuscript; imported, not proof-audited here.

[3] A. Smith, *The Birch and Swinnerton-Dyer conjecture implies Goldfeld's conjecture*, 2025. arXiv:2503.17619.

[4] P. Koymans and A. Smith, *Tamagawa ratios and unbounded Selmer moments*, June 30, 2026, Theorem 1.4. arXiv:2606.31649v1.

[5] X. Li, *Moments of quadratic twists of modular L-functions*, Invent. Math. 237 (2024), 697–733. arXiv:2208.07343; publisher DOI.

[6] E. Kowalski, P. Michel and J. VanderKam, *Non-vanishing of high derivatives of automorphic L-functions at the center of the critical strip*, J. Reine Angew. Math. 526 (2000), 1–34. Author-hosted manuscript.

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[8] D. Fiorilli, *A conditional determination of the average rank of elliptic curves*, J. London Math. Soc. 94 (2016), 767–792. arXiv:1403.7108.

[9] N. Creighton, *Upper Tail Distributions of Central L-values of Quadratic Twists of Elliptic Curves at the Variance Scale*, Quarterly J. Math., published October 7, 2026, haag040. Publisher DOI.